

■ Re-Grow: 4-Minute Demo Video Teleprompter & Production Script

Complete Video Creator Guide: Exact Voiceover Timings, Screen Mouse Choreography & Text Overlays

Pre-Recording Setup Checklist	Target Setting / Action
Screen Resolution & Browser Zoom	1080p (1920x1080) at 110% Browser Zoom for maximum legibility.
Local Server Execution	Run python run.py in terminal. Verify API status badge is Green (Online) .
Microphone & Audio Settings	Use clean noise-canceling mic. Pace speech at ~130-140 words per minute .
Browser Mouse Pointer	Enable smooth cursor highlighting. Avoid erratic mouse jitter during recording.

■ SCENE 1: The Crisis & Hook (0:00 - 0:35)

■ **SCREEN ACTION:** Start on Re-Grow Hero page (<http://localhost:3000>). Slowly pan down over headline & metric pill badges.

■ **ON-SCREEN TEXT OVERLAY:** DISASTER MONITORING IS BROKEN → RE-GROW: FROM PIXELS TO SAPLINGS

■ **EXACT VOICEOVER SCRIPT:**

"Every single year, over 10 million hectares of forest are destroyed by deforestation and intense wildfires. Current disaster tools only show satellite maps of *where* trees died. But passive maps leave recovery teams stranded—with zero actionable budgets, zero sapling counts, and no local ground partners. Meet Re-Grow—an Active Disaster Recovery Platform that transforms raw satellite pixels directly into budgeted, actionable restoration plans in under 5 seconds!"

■ SCENE 2: Live Sentinel-2 STAC Acquisition (0:35 - 1:15)

■ **SCREEN ACTION:** Click ■ **Live Sentinel-2 Stream** tab. Type 'Amazon Rainforest' in search bar, click **Locate**, then click **Fetch Cloud-Free Satellite Data**.

■ **ON-SCREEN TEXT OVERLAY:** LIVE ESA SENTINEL-2 STAC API: Automating Cloud-Free Tile Fetching

■ **EXACT VOICEOVER SCRIPT:**

"Responders can upload custom drone GeoTIFFs, or use our Live Sentinel-2 Stream. I'll search 'Amazon Rainforest' and click Locate. Our Python FastAPI backend connects to Microsoft Planetary Computer STAC API across European Space Agency Sentinel-2 satellites. It automatically filters for cloud-free tiles and loads baseline historical imagery alongside recent post-disaster satellite tiles. Now, I'll click Execute Deforestation Analysis!"

■ SCENE 3: Computer Vision & AI Damage Heatmap (1:15 - 2:00)

■ **SCREEN ACTION:** Page smooth scrolls to **Comparison Slider**. Drag handle left/right. Toggle **AI Mask Overlay**. Click **Download Overlay PNG**.

■ **ON-SCREEN TEXT OVERLAY:** COMPUTER VISION ENGINE: Normalized Greenness Index (ΔG) + Adaptive Thresholding

■ **EXACT VOICEOVER SCRIPT:**

"Our computer vision engine computes normalized greenness index deltas with adaptive statistical thresholding to isolate deforested clusters. On our Interactive Satellite Comparison Slider, responders can drag the divider to inspect forest loss pixel-by-pixel, or switch to AI Mask Overlay to view detected damage highlighted in red. We can even download high-resolution visualization PNGs instantly!"

■ SCENE 4: Reforestation Economics & Live Telemetry (2:00 - 2:45)

■ **SCREEN ACTION:** *Hover over **Live Climate Telemetry Card** (Temperature, Wind Speed, Fire Spread Risk) \$ ightarrow\$ Hover over **KPI Metrics Grid Cards**.*

■ **ON-SCREEN TEXT OVERLAY:** REFORESTATION ECONOMICS: 1 Tree / 4m² | \$2.50 per Sapling Benchmark

■ **EXACT VOICEOVER SCRIPT:**

"Above our metrics, Re-Grow streams real-time environmental telemetry from Open-Meteo—monitoring temperature, wind speed, and fire spread risk.

Here is the core breakthrough: Re-Grow detected 6,018 square meters of lost forest. Applying global ecological density standards of 1 tree per 4 square meters, it calculates an exact requirement of 1,504 saplings and an estimated recovery budget of \$3,761.25!"

■ SCENE 5: Geospatial NGO Matchmaker & CSV Export (2:45 - 3:30)

■ **SCREEN ACTION:** *Scroll to OpenStreetMap location pin \$ ightarrow\$ Click **Export CSV** on Regions Table \$ ightarrow\$ Hover over **NGO Partner Cards**.*

■ **ON-SCREEN TEXT OVERLAY:** GEOSPATIAL MATCHMAKER: OpenStreetMap Overpass API 50km Query

■ **EXACT VOICEOVER SCRIPT:**

"Money alone doesn't plant trees—you need local boots on the ground. Re-Grow queries OpenStreetMap Overpass API to find registered forestry offices, nature clubs, and environmental non-profits within 50km of the damage.

Responders can click Connect & Partner to visit verified websites, export cluster bounding box data via Export CSV, or view past runs in our SQLite database!"

■ SCENE 6: Tech Architecture & Conclusion (3:30 - 4:00)

■ **SCREEN ACTION:** *Show Next.js 15 + FastAPI architecture diagram / GitHub repo screen \$ ightarrow\$ End card with Re-Grow branding.*

■ **ON-SCREEN TEXT OVERLAY:** Re-Grow: Next.js 15 | FastAPI | Sentinel-2 | OpenStreetMap

■ **EXACT VOICEOVER SCRIPT:**

"Built with Next.js 15, FastAPI, OpenCV, and OpenStreetMap, Re-Grow turns satellite pixels into real saplings planted. Thank you for watching, and let's re-grow our planet together!"